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Social equity indicators for a Blue Economy: Guidelines for application

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ABSTRACT

Social equity goals are specifically stated in most new ocean management and development plans, notably those aligned with a Blue Economy that recognizes the significant and rising inequities across ocean spaces and sectors. However, the deeper significance of social equity has still not been fully integrated in policy and management practices, which often ask why it is relevant and how it can be measured in ways comparable to more familiar ecological and economic indicators. We propose a theoretical and practical framework (including examples of indicators) that builds on key concepts in critical environmental justice, antiracist and decolonial scholarships: difference (not all groups have equal opportunity, so interventions must serve the least well-off); intersectionality (people and groups have multiple self-identities); power relations (colonization has entrenched an 'us-vs-them' mentality); and interest convergence (goals of marginalized groups are often only pursued if they benefit the powerful). These principles are contrasted with three categories of indicators that sequentially build toward evaluating social equity goals: total outputs (what and how much do ocean sectors contribute or impact); disaggregated impacts (who shares in specific benefits and costs); and equity actions (specific actions taken to implement equitable processes and outcomes). Local, national, and intergovernmental evaluation plans can use these guidelines to clarify what aspects of development are being reflected in indicators, and what indicators need to be added for a fuller picture of outcomes that support the interests of marginalized populations across the world's oceans.

1. Introduction

The last decades have seen a major shift in the direction and scope of ocean policy as its contributions to people—and the scale of our impacts—are better understood. The 'Blue Acceleration' in ocean industries such as mariculture, bioprospecting, and renewable energy [12, 41, 57]—has rapidly increased new economic opportunities but done so in ways that benefit only a few. This does not only exclude people from benefiting from new industries, but often actively dispossesses and marginalizes local communities from access to their traditional territories and resources [10, 12, 21, 44]. A focus on sustainable economic growth is therefore not enough to deliver truly equitable development, and could be counterproductive given the political nature of benefit distribution and the unequal power structures embedded in all aspects of policy development and implementation [49, 51]. Addressing this

requires new ways of understanding social benefits and new priorities to select the indicators we use to monitor progress [27, 49, 67].

This paper builds on research into multidisciplinary indicators, which has highlighted a particular need for ways track progress on equity and justice goals in marine sectors [18, 45, 54], but has not yet explicitly connected recommendations from social justice scholarship and civil rights movements to help choose appropriate indicators. We argue that filling this gap is necessary to avoid equity considerations becoming simple checklists in Blue Economy implementation, and, rather, for plans and projects to be grounded on a clear understanding of critical approaches to acknowledge and address underlying social interactions and marginalization in specific contexts. Thus, we propose using established principles addressing class, race, gender and colonial injustice to guide the design of Blue Economy indicators through the knowledge and scholarship of social justice and civil rights movements

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beyond ocean-specific research. We focus here on selecting and using indicators, as there are a number of reviews of broader conceptualizations and integration of equity in ocean science and policy (see, among others, [1,9,28,48,49]).

The need for this set of principles builds from three key assumptions considered throughout the paper. First, social equity—for ethical and instrumental reasons—is at the foundation of the Blue Economy as a transformative approach and distinguishes it from unsustainable and marginalizing ocean development [20]. Second, the tracking of indicators related to policy process is as important as indicators related to policy outcomes. Third, while some general indicators can be useful across cases, successfully planning and implementing local blue economies requires a critical examination of their specific context, which necessarily involves the feedback and leadership of frontline communities—that is, those directly involved in and impacted by the Blue Economy.

We expand on the concept of social equity in the following section but generally use the term in the context of ocean governance, especially to inform impact assessment of ocean sectors, to promote inclusive community engagement in ocean management, and to enable transformative change in ocean policy. This concept of social equity in ocean governance is highly relevant to recent multilateral agreements, and commitments have highlighted the vital importance of cooperation across the political and economic spectrum. These include the FAO's Voluntary Guidelines for Small-Scale Fisheries [17,33], the WTO Agreement on Fisheries Subsidies [22], the inclusion of a distinct target for artisanal fishing benefits in the UN 2030 Agenda (SDG 14.a), and the designation of 2021–2030 as the UN Decade for the Ocean and 2022 as the International Year of Artisanal Fisheries and Aquaculture. A through-line in these commitments is a need for transformative change—"a fundamental, system-wide reorganisation across technological, economic and social factors, including paradigms, goals and values" [40]—so social equity must be central to the pursuit of a Blue Economy [27].

In that context, it is important to emphasize that a focus on social equity was indeed intended to set the Blue Economy apart from pure economic quantification of the ocean economy, or more narrow growth-oriented industrialization goals of 'blue growth' that undermine environmental sustainability [20]. However, the term 'Blue Economy' is now used by many different actors and under a correspondingly wide set of emphases [61,76], and is even viewed skeptically by some marginalized groups that the proponents of its original meaning intended to support [15]. As will be discussed at length in the following sections, the perception of how and who benefits from ocean development depends on the indicators we use, with policy and management strategies relating to the blue economy following from the resulting identified priorities.

Here, we first provide a brief background to the Blue Economy in the context of emphasizing social equity-focused economic development, including procedural and distributional aspects. We then undertake a more in-depth discussion of the equity and justice principles that underpin the consideration and selection of equity-focused indicators. This encompasses a broad and significant volume of scholarship; however, here we narrow the focus specifically to its importance for marine policy indicators. Finally, we discuss benefits and examples that transition from absolute outcomes to disaggregated outcomes to process indicators. Throughout the discussion, we highlight guidelines that can be followed in many different cases to select such indicators but emphasize that their selection and use has to be decided for specific contexts.

2. An equity-focused Blue Economy

The notion of social equity is increasingly a key goal in development, and this is reflected by its inclusion into existing frameworks and the creation of more specific ones. One example is the UNEP Sustainable Blue Economy Transitions Framework (BETF), which is intended to help

nations transition to blue economies along the five guiding principles of i) healthy ecosystems, ii) equity and inclusivity, iii) climate stability and resilience, iv) sustainable consumption and production, and v) circular processes [27,72]. This is not a straightforward process, yet it must always involve carefully considering what is meant by social equity, whose perspectives and needs are being marginalized, what current equity frameworks can be leveraged, and how social equity issues can be meaningfully addressed to legitimize policy processes, redress historical harms, and ensure fair outcomes in the future.

Following from this, Croft et al. [27] proposed a model for incorporating social equity within blue economies by defining goals and objectives that are centered on four main principles drawn from key research in the field (see, e.g., [6,28,43])—Achieving Fairness, Preventing Harm, Respecting Rights, and Supporting Flourishing. "Achieving Fairness" refers to an equitable distribution of costs and benefits from ocean sectors and is closely linked with the use of disaggregated indicators as will be discussed below. "Preventing Harm" includes avoidance and mitigation but also compensation when harms do occur. "Respecting Rights" extends to both formal and informal expectations and practices held by groups and individuals. Many traditional rights and practices have indeed been suppressed or ignored through processes of colonialism and discrimination, so in addition to respecting rights, the *restoration* of rights is a necessary first step requiring equity-enhancing actions [75]. "Supporting Flourishing" builds on these equity principles to raise the bar for development goals, from providing access to basic rights such as food, health, and employment to further enabling well-being in all aspects [27], commensurate with the UN Sustainable Development Goals [71].

In order to achieve these goals, related research points to a need to adopt anti-inequity and anti-racist actions—these are actions that explicitly avoid processes and outcomes that marginalize individuals and groups on the basis of their identity or ascription—as a way to dismantle and disrupt historical inequitable systems in science and policy planning and implementation [2,37]. This is important because equity outcomes can perhaps be tracked but equity goals cannot be achieved within existing systemic discrimination and abuse [49,65]. Therefore, the indicators that have been historically used to track implementation and outcomes in ocean development may often not be appropriate for a Blue Economy approach and must be carefully re-considered to actively work to achieve equity goals and transform both established and emerging sectors.

2.1. Key principles for indicators in an equitable Blue Economy

Indicators are often proxies for concepts that are multidimensional, context-specific, and difficult to measure directly. As highlighted above, social equity is a particularly complex goal but can be defined in practice [27,28]. To aid in the selection of indicators for an equitable Blue Economy, from total outputs to disaggregated impacts to equity actions (discussed at length in Section 4), in this section we highlight four principles (Difference, Intersectionality, Power Relations, and Interest Convergence) based on fundamental and cross-cutting aspects of social (in)justice and subordination, such as class (economic), gender, colonization and race. These have been considered throughout moral philosophy, feminist ethics, anti-colonial studies and critical race theory, and have been previously highlighted with respect to ocean-related policy (see, e.g., a review by [65]). We discuss these particular principles here as they apply in practice to design actions that require disaggregated impact indicators rather than total outputs, in order to be specific to the social contexts of each common type of subordination (class, gender, colonial, racial). Total output indicators are thus only a first baseline that must be disaggregated to address principles of difference and intersectionality; resulting disaggregated impact indicators can then be examined through power relations and interest convergence principles to determine meaningful equity actions and their indicators (Fig. 1). We explain and discuss each of these principles at length below

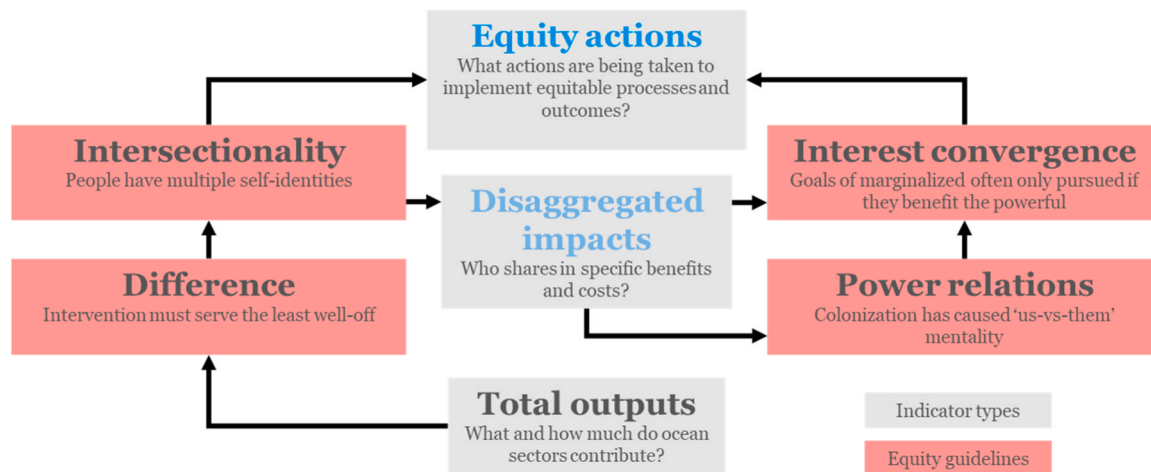


Fig. 1. Key equity and justice principles to guide selection of indicators focusing on total outputs, disaggregated impacts, and equity actions in a Blue Economy (Table 1).

and then provide specific examples of indicators within different categories.

2.2. Difference

The Difference principle holds that equal liberties and opportunities must be attained, and equal opportunities to contribute to decisions be granted to ensure equitable decision-making and outcomes. This is similar to the concept of anti-subordination that seeks to specifically identify and prevent practices that subordinate a specific identity group [25]. Once this is in place, this principle draws attention to the justice and fairness of interventions that serve those least well off [55]. The difference principle can be augmented by the *capabilities* approach [3, 59], which stresses that the goal of policy initiatives should not be only to improve opportunities, freedoms and resources for the worst-off people in the community, but to ensure that those goods are facilitating people's actual choices and activities (their capabilities) in ways that they value. Any treatment of equity has to take into account historic injustices that shaped contemporary circumstances and acknowledge the accumulated advantage that some groups have gained over others due to systemic biases and overt oppression. In practice this means a focus on disaggregated data which identifies those who face the most inequity procedurally and in outcomes. In this sense, considering these disaggregated impacts represents an essential first step in efforts towards equitable processes and outcomes in a Blue Economy.

2.3. Intersectionality

How to identify those "left behind" is not trivial. We therefore suggest the second guiding principle to follow the principle of intersectionality [11,26]. Intersectionality holds that people hold multiple identities and applying policies that differentiate people too coarsely will fail to appreciate the kinds of systematic pressures that different groups face. It also runs the risk of forming policy that caters to the dominant group and rendering priorities from underrepresented groups as invisible. In operationalizing the principle of intersectionality, we suggest that any options for disaggregated indicators are considered through a range of possible dimensions of inequity (e.g., gender, race, age, ethnicity, nationality, etc.; see Table 1).

These different dimensions of systematic inequity are not mutually exclusive but also not fully substitutable. Considering race and ethnicity means thinking about which racial/ethnic groups have faced limited opportunities or been historically oppressed in favour of dominant groups [47,5]. Considerations of class should focus on which socioeconomic groups (either through hereditary titles, monetary inheritance, or

purchasing power) have historically benefitted from policy and influence it in their interests. Gender considerations must recognize that in most cultures, there have been (and continue to be) differential impacts of policy and differential roles in leadership between genders [16].

Colonial histories have reshaped social hierarchies and notions of how society should be structured, usually suppressing the philosophical, political, and cultural practices of the colonized groups. Even notions of equity itself can be shaped by colonial histories [1]. Political power recognizes that in every decision context, some groups have a greater role in decision-making than others. Often this can follow generalized colonial legacies, but in some cases it may not. For example, the USA is a former colony but in international diplomatic settings often has an outsized role in decision-making that other former and current colonies do not possess. In filtering a selection of indicators through these considerations, we argue that the resulting chosen indicators will more accurately identify the target demographics to base our measures of equity.

2.4. Power relations

The third principle follows the concept of power relations, a central concept in post-colonial studies that holds that our understanding of the world is through the lens of the global West and North. Most contemporary academic institutions, individuals, and works are western and legacies of European colonial powers, and this influences the types of knowledge and individuals that inform decision-making [67]. Acknowledging that any academic, philosophical, scientific, or artistic pursuit can never be truly objective, because research priorities and perspectives on information are interpreted through specific cultural lenses [64], this means that our understanding of the world may not reflect the understanding of many of the people who inhabit it [68]. Crucially, this principle holds that much of what we understand of the global East and South are social constructs and impositions of European academic legacies. Colonial legacies have therefore set up binary power relationships which describe an us-and-them dynamic of understanding the world, where the global East and South are often the objects of study and less likely to have their priorities realized in research and policy, much less their interpretation of the world [46].

To conduct assessment in recognition of this principle, we suggest that any selection process of indicators must include representation from the groups that equity-serving interventions are supposed to serve. Neglecting to include these groups in a co-produced process of prioritization risks selecting indicators that do not reflect the priorities of the groups whom interventions are meant to benefit. Engaging with these groups will often require capacity-building or enhancing measures to

Table 1

Worked examples of indicators for tracking progress on social equity goals in a Blue Economy, with descriptions of indicator types including Total Outputs, Disaggregated Impacts, and Equity actions.

Indicator Category	Indicator examples
Total Outputs <i>How much?</i> Total contribution of ocean sectors to economies at particular scales. This can also include effects of ocean sectors on ecosystems (these are impacts on natural capital).	<i>Contributions to well-being</i> ■ Fisheries catch, landed value, stock status. ■ Aquaculture production, trade value. ■ Fishworkers employed (full or part-time). ■ Tourist expenditures and employment generated by marine ecotourism. ■ Revenue received from blue carbon credits. ■ Energy generated offshore. <i>Impacts on ecosystem services and natural capital</i> ■ Number of overfished stocks. ■ Water pollution from coastal activity and/or agriculture. ■ Restoration of habitat quality for nutrient cycling. ■ Repopulation of local species. ■ Re/De-forestation of coastal vegetation. ■ Space used by offshore facilities.
Disaggregated Impacts <i>Who and where?</i> Similar metrics to total output indicators, but specifically collected for different groups of people.	<i>Contributions and Impacts by groups based on (e.g.):</i> ■ Ethnicity ■ Gender ■ Race ■ Geographic area ■ Indigeneity ■ Internal/Foreign Migrants ■ Language ■ Modes of practice (e.g., artisanal, industrial) ■ Nationality ■ Political affiliation ■ Wealth/Class ■ Traditional ownership
Equity Actions <i>What is being done?</i> Initiatives and actions taken to improve equity outcomes and processes.	<i>Following from identified groups above:</i> ■ Financial support, by group. ■ Conferral of ownership in sectors, by group. ■ Delegation of management decisions, by group. ■ Participation in earliest stages of planning, by group. ■ Hiring of planners and managers, by group.

allow them to take part in these decision processes, and the utilization of social scientists who are trained in facilitating and eliciting perspectives of groups of people [60].

2.5. Interest convergence

As a mirror to the third principle, the fourth and final principle is that of interest convergence [8]. Interest convergence recognizes that gains for marginalized groups usually only progress when they align with the interests of groups with the power to affect policy and lead interventions. Of particular concern are cases where powerful groups may want to minimize change to their modes of practice and focus on how they promote a perception of equity. That is, if a powerful proponent of an intervention can make symbolic gains to equity (or focus instead on intentions to promote equity rather than on equity outcomes) with little

to no substantive gains felt by the people who are supposed to be supported by interventions, then the proponent can gain in reputation and succeed in operationalising interventions that are equitable in name only. These effects are clear across ocean spaces, where (neo)colonial decision-making did not consider local benefits unless they were beneficial for the metropole, with negative effects on development and capacity persisting for centuries afterward [23]. This continues to occur, for example, when observing outsized influence of developed-nation goals in voluntary commitments for ocean actions (to the detriment of local goals in formerly-colonized nations) or in Blue Economy funding streams nominally intended for the global good but ultimately providing most funds to developed nations [57].

Therefore, to guard against the risks that interest convergence presents, we suggest operationalizing this principle by limiting the role of external proponents of interventions in selecting and monitoring indicators (unless the proponent is a currently excluded community). Note that the tracking of such roles is itself an indicator of process equity actions; following from the points above on power relations, communities engaged with a Blue Economy should themselves establish the standards by which this would be monitored. There should be checks against potential conflicts of interest whereby a proponent monitors its own performance, as monitoring equity should be in the service of the group who should gain from equity and not the group promising it through an intervention in their own interest [60]. Therefore, we recommend that capacity-building measures are provided for the affected community to monitor equity or have a third party in charge of monitoring equity throughout an intervention.

3. Indicators for an equitable Blue Economy: from total output to equity action

There is an enormous array of possible indicators (and selection frameworks) on economic, social, and ecological aspects relevant to ocean development (see, e.g., [18,24,50]). As discussed above, however, meaningfully integrating equity in blue economies requires that planners specifically anticipate how their choices of indicators and evaluation methods perpetuate or counteract systemic inequity in outcomes and processes (Fig. 1). Comparing relative outcomes across groups, places, or time, can highlight differences in some aspect of wellbeing, but can easily frame equity in the context of dominant groups and neglect the perspective of marginalized groups. For example, some Blue Economy discussions have appropriately emphasized women's employment in ocean sectors, which is essential for supporting labor rights and their inclusion in decision-making but still follows a largely Eurocentric notion of the value of work within a market economy that marginalizes 'non-productive' labor and cultures [36,42]. Another concern with comparative goals is that some interventions may increase comparative inequality at the same time as they improve the absolute position of the worst off. Often, it is this latter improvement (the conditions of the worst-off group) that is crucial for equity and fairness [55, 7].

Whether using absolute or relative indicators, tracking policy progress in general can follow one of four approaches based on comparing current states to 1) a threshold (whether an ideal or current standard); 2) past conditions; 3) a proxy (assumed to be directly correlated with the ultimate goal); or 4) a counterfactual condition (i.e., expected outcomes without an intervention). All four kinds of measurement are useful and are often done together, but can also be misleading and misused.

Comparisons to a threshold can be meaningful where there are specific measurable objectives, such as drinking water quality (e.g., microbial loads). People without access to drinking water is a clear sign of inequity. However, only defining success against an ideal standard may detract from observable progress towards that standard. Comparing to past conditions can thus help appreciate the effects of an intervention but can also lead to misleading conclusions if other factors are also influencing change or if past conditions are an inadequate baseline for

progress. For example, small increases in access to food in communities that are still chronically malnourished could hardly be considered a success. Comparing to a proxy is commonly used when ongoing measurements on the subject of interest are not feasible or the goals themselves are conceptual (e.g., happiness or fairness). However, care must be taken to ensure that a proxy is meaningful for the specific goal. For example, access to project financing is a common proxy indicator for policy progress but may or may not lead to observed change or address local priorities. Comparisons to a counterfactual—a hypothetical scenario if the intervention had not occurred—are difficult because they require credible ways of projecting past trends and likely effects. However, if possible, it would be the best way to understand the specific effect of a given intervention.

Indicators can be divided into three main categories as one transitions from pure quantification of an ocean economy to a transformational approach under a Blue Economy: 1) Total outputs, 2) Disaggregated impacts, and 3) Equity actions (Fig. 2). These broadly map onto the four dimensions of equity discussed above [27]. Of course, any indicator must be considered specifically for a given case. Each category of indicator is summarized in Table 1 and discussed at length in the next sections. Total output indicators broadly cover Preventing Harm and Supporting Flourishing; disaggregated impacts reflect Achieving Fairness and Respecting Rights; and equity actions reflect processes for all four equity types.

3.1. Total output indicators

Total outputs refer to the overall contributions of ocean sectors to the economy. These include classic metrics such as total fisheries catch, aquaculture production, employment in coastal sectors, and revenue from ecotourism, among others. However, this also includes ocean and coastal ecosystem services and natural capital that may be positively or negatively affected by human actions and that may or may not be obtained through or traded in market economies (Table 1).

Because equity is multi-dimensional and cumulative [27,28], indicators on total 'outputs' should be expanded beyond market benefits to include concurrent trends in underlying ecosystems and communities [19,70] (Table 1). By combining indicators on these two aspects of ocean development (production and impact), one can create more useful indices of efficiency, productivity, and sustainability. For fisheries, for example, it has long been recognized that an indicator of catch per unit of effort (i.e., accounting for historical increases in fishing power) can sometimes be more informative for policy than simply tabulating catch. In emerging sectors such as ocean energy, indicators should similarly

track energy production (for example, in gigawatts) compared to investment costs in offshore wind or tidal installations compared with the costs of alternative forms of energy production, but also accounting for possible environmental impacts (of course, environmental costs should be considered when evaluating any form of development).

While total output indicators are generally quite coarse, the general lack of data on economic dynamics in many ocean sectors means that these indicators may sometimes be the only form of standardized data collected in a country or region. That being the case, they have historically been used as implied proxies for social well-being, for example by assuming that there are population-wide gains (or losses) in seafood supply or income when ocean sectors grow. Indeed, these total output indicators can be complemented with local-based understandings of social processes and events affecting ocean sectors to provide deeper insights into community well-being and can sometimes flag emerging issues. Clearly, however, this is insufficient to adequately gauge the effects of new policies and their strategies, particularly when the distribution of impacts is a primary focus.

3.2. Disaggregated impact indicators

Disaggregated impact indicators specifically distinguish who shares in benefits and/or bears costs from ocean sectors (Table 1). This also considers where benefits are produced (e.g., where a fish was caught) compared to where they are enjoyed (e.g., where it is eaten, or where value is added). In the context of equitable development, this is an absolutely vital improvement over total output indicators.

Some disaggregated impact indicators can be relatively simple, such as the share of fisheries catch (or revenue, etc.) of artisanal versus industrial fleets, or between two nations sharing a stock or operating in a transboundary area [29]. This could also include the most well-known indicators of inequality, such as the Gini coefficient (the share of benefits held by each percentage of a population). They can also be more specific, for example including recent efforts to better recognize and account for the participation of women in the seafood sector [39], or to elicit and highlight labor abuses occurring in ocean sectors, usually disproportionately impacting immigrant communities and individuals [31]. The collection of disaggregated impact indicators is a vital step towards achieving more equitable ocean sectors and can prompt important discussions regarding the specific groups to be monitored in a given setting.

Disaggregated impact indicators are important because historical development is vastly inequitable; corporate and national consolidation of resources are stark, pervasive and concerning in both established and

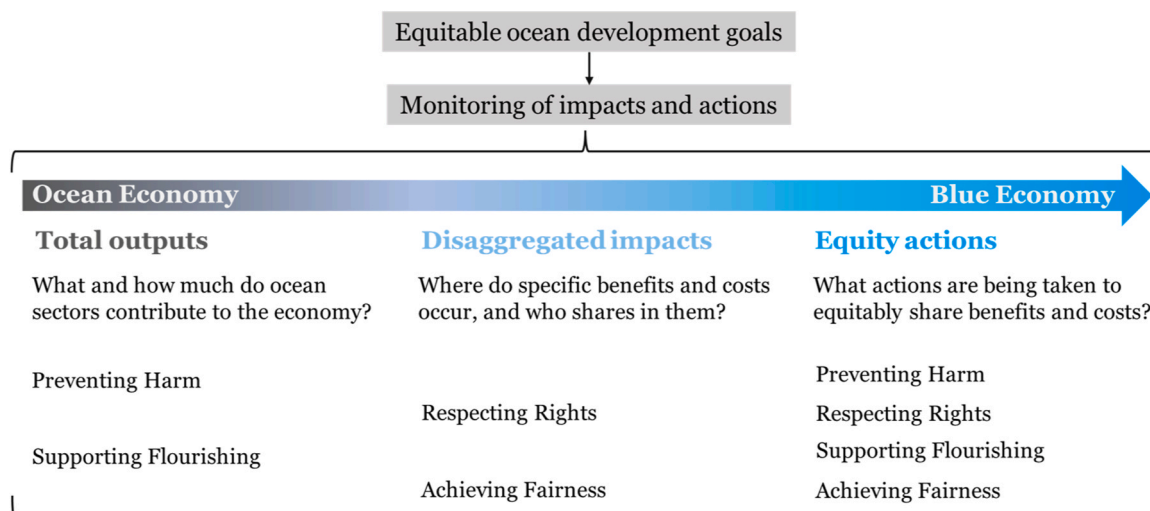


Fig. 2. Categories of indicators for transitioning from Ocean Economy to Blue Economy development in marine and coastal sectors.

emerging sectors [66,74]. For example, government financial support for large industrialized fisheries is twice as much relative to their landed value as that for small-scale fishers [56], while 97 % percent of marine genetic resources have been patented by companies headquartered in just 10 countries [12]. Mechanisms that can perpetuate inequities from the ocean economy include insecure territorial and tenure rights, historical and colonial legacies of marginalization from resources and their benefits, and lack of access to resources, financial and technological capacity [14,23]. All of these differences in the distribution of resources, benefits, and impacts can occur between different nations [29], but also across gender, racial, ethnic, and cultural divides. This underscores the marked differences in perceptions and appropriate policy that can arise from a relatively simple yet more meaningful disaggregation of total output indicators.

3.3. Equity action indicators

Equity action indicators track what is being done to support equitable processes and outcomes, and to combat existing systemic or individual inequities; these are anti-inequity and anti-racist actions. In order to truly advance towards an equitable Blue Economy, indicators related to equity actions, such as anti-racism and decolonization, are an essential next step to disaggregated indicators because they focus on the specific initiatives being taken to advance equity in a specific context. In this sense, while output and disaggregated indicators offer increasingly nuanced *descriptions* of outcomes (and inequalities and possible inequities), indicators of equity actions track progress on our policy responses to address identified equity issues.

Inequities are the result of historic and contemporary processes that are part of current structures of governance that are evident both in national settings and across entire regions. In the Caribbean, for example, the current overall lack of governance and infrastructure capacity to establish a Blue Economy can be directly traced back to national colonial and neocolonial histories [23]. The extent of such structural inequities requires active efforts to counteract, so it is important to recognize that the data we use to monitor inequities (the indicators) are often not the prime interest of equity goals. Rather, what is of interest is to understand and reflect on the processes that generate the data that inform subsequent decisions. That is, any policy or intervention program designed to address or promote equity should clearly connect indicators with procedural (or causal) links with the processes contributing to inequities. How do policies and interventions act to stymie these processes? Do the policies and interventions instead risk enhancing inequities through these processes? There is a clear need for alignment across actors and institutions at different scales in addressing these issues [63], but improving equity outcomes must involve bottom-up leadership and thus equity actions can focus on supporting access and tools to lead.

For example, decision-making processes as part of Regional Fisheries Management Organizations (RFMOs) have been criticized for involving systemic inequities in the form of unequal bargaining on fishing access allocations between members with differing capacity to negotiate or even attend meetings [52]. In the context of our discussion, this is compounded by the use of indicators that consider disaggregated catch by fishing nation and not by where fish were caught. This places higher weight on nations with large fleets, to the detriment of nations with small or no fleets but whose EEZs actually sustain the fish stocks [29]. In this case, a total output indicator (say, total catches or employment) would offer some insights on sustainability or economic performance, disaggregated impact indicators (catch by nation) may show inequalities, yet more meaningful equity actions and their indicators (changes in membership or in allocation processes) would require a concerted effort to understand power relations and interests within RFMOs and member nations.

In settler colonial nations, for example, equity actions currently seek to increase fishing access and territorial decision-making for Indigenous

Peoples, with progress indicators including financing amounts to support purchases of quota and gear and outcomes indicators include the share of fisheries employment and resource access controlled by Indigenous-owned companies. These and similar actions throughout the world are the result of many decades of Indigenous-led struggles for recognition and their ongoing oversight of outcomes to inform policy changes, and it is these equity actions and not indicators of inequality that create change. However, this perhaps supports but does not fully address the need for deeper decolonial actions and political struggles for self-determination beyond participation [13,30,69].

Tracking financial support for equity-focused projects can itself be a useful indicator of equity actions, and work along these lines has indeed revealed global-scale patterns that prioritize large-scale projects in highly-developed nations over local-scale initiatives in developing regions [57]. Research and planning that uses finance as an indicator should be aware, however, of three pitfalls of such indicators. First, in practice it can be difficult to fully distinguish between equity-oriented projects and more general components of co-management or other collaborative efforts. To be clear, co-management and active consultation can be a pathway to foster more inclusive outcomes and are examples of equity actions that should be embedded into any new project or existing sector [21]. However, truly equity-focused actions should track progress on the quality and breadth of engagement and, ideally, have fundamental equity goals beyond industrial development.

A second issue with tracking finance is that sources aside from government or intergovernmental institutions, including private and non-governmental organizations, are often not public or otherwise less transparent regarding beneficiaries or outcomes [57]. However, a Blue Economy necessarily entails use of national ecosystems, and so the entire public—and coastal populations as a bare minimum—should be aware of funding into planning and development of any areas. Third, and perhaps most important, financing indicators must be complemented with indicators on engagement, participation, and input into policy by all of the groups identified as affected in order to avoid the inequity issues of total output indicators. (e.g., Table 1).

3.4. Selecting indicators: assumptions and linkages

Our indicator framework is focused on supporting equitable outcomes from a Blue Economy, and it is important to consider this within broader critiques of ocean development. Despite optimistic narratives regarding the overarching benefits of ocean industrialization, there is clear evidence that past (e.g., expansion of fisheries) and ongoing (e.g., seabed mineral exploration, ocean renewable energy) ocean development is characterized by highly inequitable processes and outcomes (socially and ecologically), and particularly for already marginalized peoples and regions [35,49,57].

In practice, these observed distribution outcomes mean that focusing on economic growth—even if done sustainably—cannot meaningfully benefit the frontline and vulnerable communities that national and international development agendas claim to prioritize [4,48,71], making it essential to follow equity-focused approaches to transform these systems. This has led to specific calls for a more open dialogue on development assumptions and a consideration of alternative pathways within a Blue Economy, which is already seen as a potential avenue for context-specific goals [58]. This includes, for example, degrowth—a planned downscaling of production and consumption to actively recognize both inequality and environmental thresholds [38]—or agrowth—the prioritization of ultimate social or ecological outcomes without constraining policies to those that would grow (or shrink) an economy [73]. While agrowth carries an allure of agnosticism, some argue that starting discussions from a degrowth perspective is more effective to arrive at transformative approaches given the overwhelmingly growth-focused and inequitable structures of existing development paradigms [38].

In alignment with these calls, the critical concepts of difference,

intersectionality, power relations and interest convergence and their corresponding investigation through different types of indicators (Fig. 1) are indeed intended to help question current development and support transformative change. As noted above, this requires alignment with organizations across strategic and tactical dimensions and necessarily involves leadership from local communities of place and practice. This is obviously important for identifying local goals and challenges, but also for better considering the co-benefits and trade-offs between different goals and their associated indicators. These linkages can be particularly complex in dynamic ocean systems [62], so considering local priorities and relationships is essential for development to be considered successful amidst a potentially complex array of indicator trends (but, see [18,32,34,53,54] for examples of guidelines on organizing and interpreting socioeconomic, ecological and governance indicator sets).

4. Conclusions and recommendations

Many common indicators of equality (such as Gini coefficients) are useful to identify disparity but ultimately reflect only the symptoms of inequity and not its root causes. To propose effective ways to address equity as a process we must effectively link long-term policies and interventions with efforts to remedy and prevent existing harms and disparities rather than only identifying them.

For an equity-based Blue Economy, indicators must evolve from quantifying total outputs of ocean sectors to showing disaggregated impacts of both benefits and costs across all different affected groups and, crucially, tracking actions that specifically promote social equity. Designing and selecting equity-action indicators can be guided in part by considering four key principles. First, interventions must serve the least-well off (difference principle). Second, individuals and groups can identify according to multiple characteristics (intersectionality principle). Third and fourth, colonization and other processes causing power disparities often lead to ‘us-versus-them’ thinking among different groups (power relations), so that the goals of the marginalized are often only taken into account if they benefit the powerful (interest convergence). The first two principles can help transition from total output indicators to disaggregated impact and equity action indicators by recognizing all of the different peoples who are affected by ocean sectors and policies. The third and fourth principles can further inform equity action indicators by helping practitioners be aware of power imbalances in selecting the best courses of action at any stage in an initiative, from conception, to planning, to implementation, to operations, to decommissioning and divestment.

The guidelines and examples of indicators presented here will hopefully help inform planning and decisions for more equitable Blue Economies across the world, but these decisions will always have to take local contexts into account. At the very least, we urge readers and practitioners to thoughtfully consider who should benefit from a Blue Economy, how they can be freed from any subordinated relationships, and how these processes and their expected outcomes can be measured most directly to ensure we are doing all we can do to protect the interest of frontline and marginalized populations across the world’s oceans.

CRediT authorship contribution statement

Cisneros-Montemayor Andrés M.: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **H. Breakey:** Writing – review & editing, Writing – original draft. **F. Croft:** Writing – review & editing, Writing – original draft. **I. Issifu:** Writing – review & editing, Writing – original draft. **S. Bossier:** Writing – review & editing, Writing – original draft. **M. Voyer:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Y. Ota:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **J. Keefer:** Writing – review & editing, Writing – original draft. **Singh G.**

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Data availability

No data was used for the research described in the article.

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